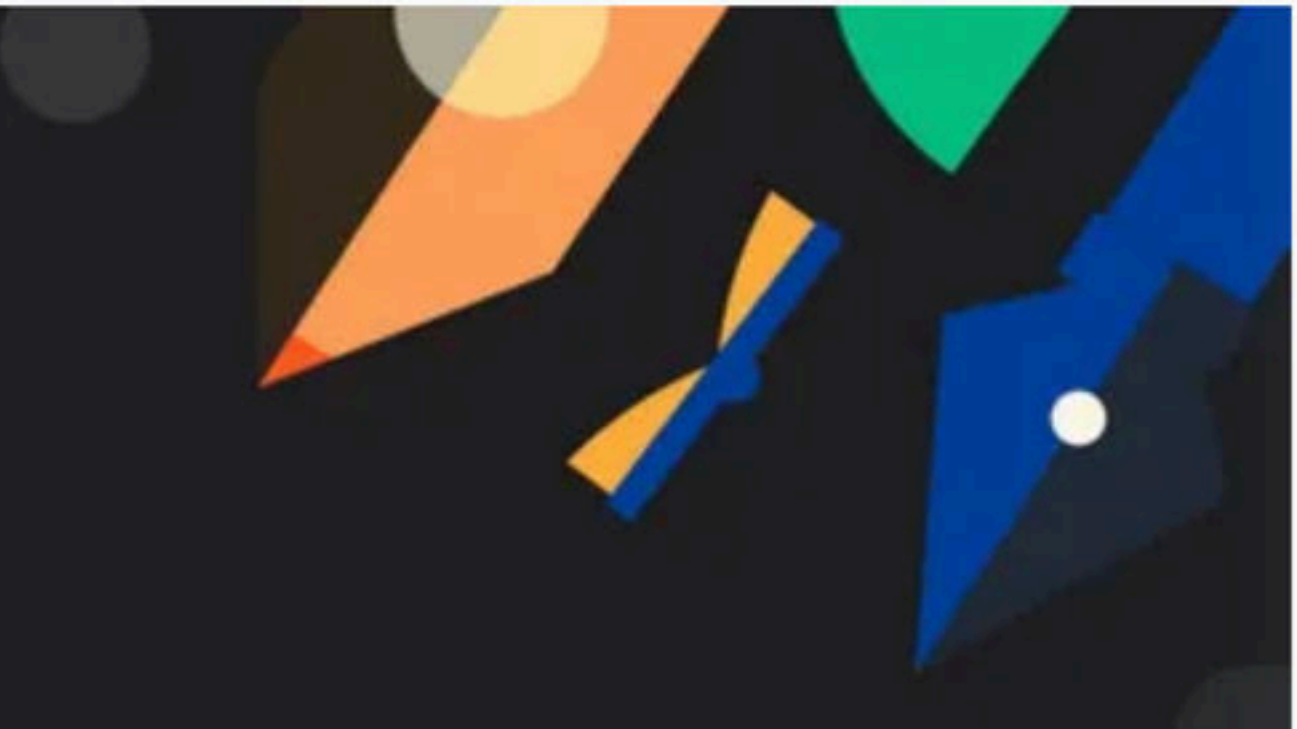




Trees Class - 1

Special class

→ Trees → D.S → non-linear





→ 0, 1, 2
Binary tree

Level 0

root

ancestor

node
↳ data
↳ next

Level 1
child of 10

sibling

child of 10

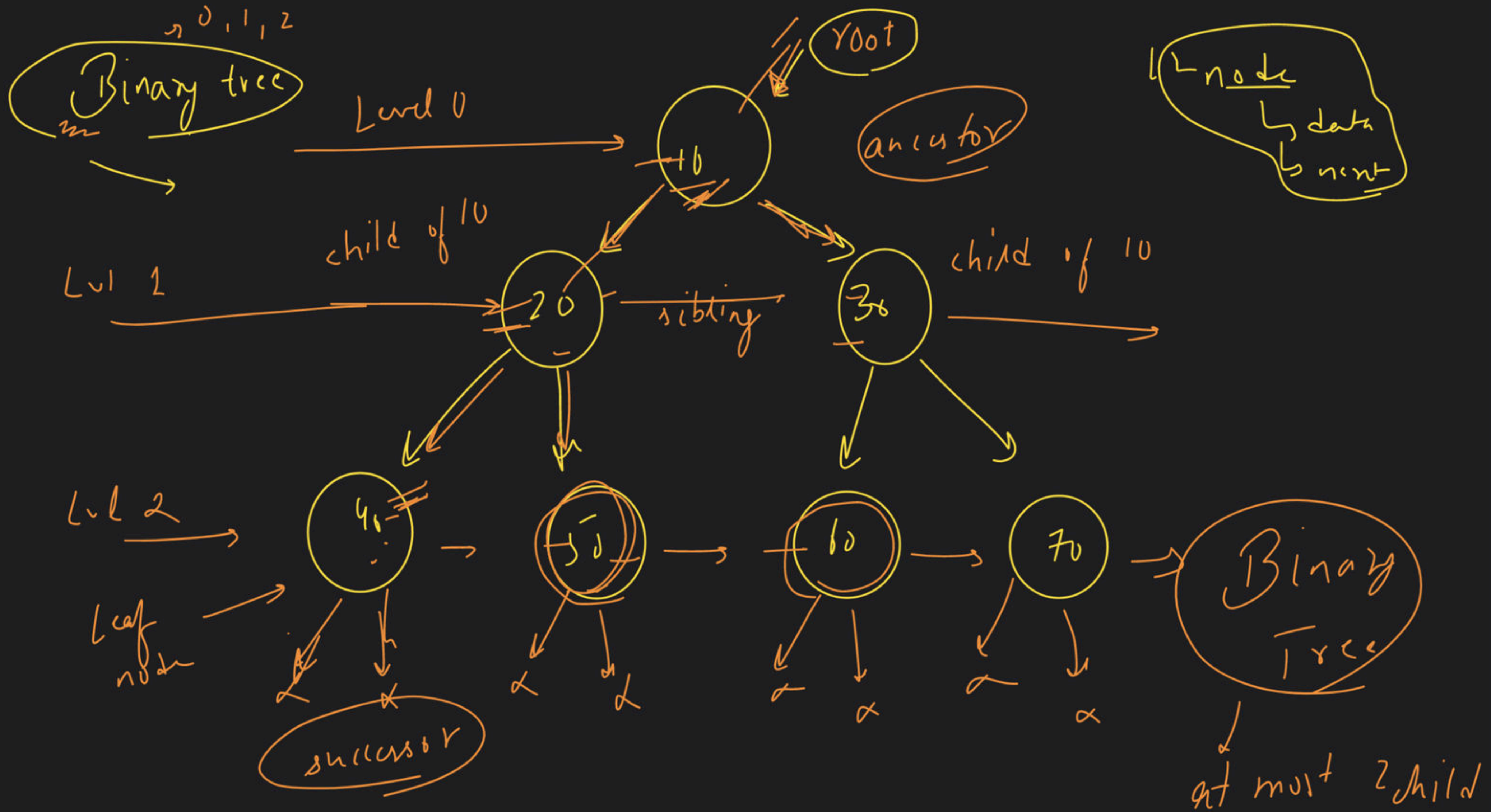
Level 2

leaf node

successor

Binary Tree

at most 2 child

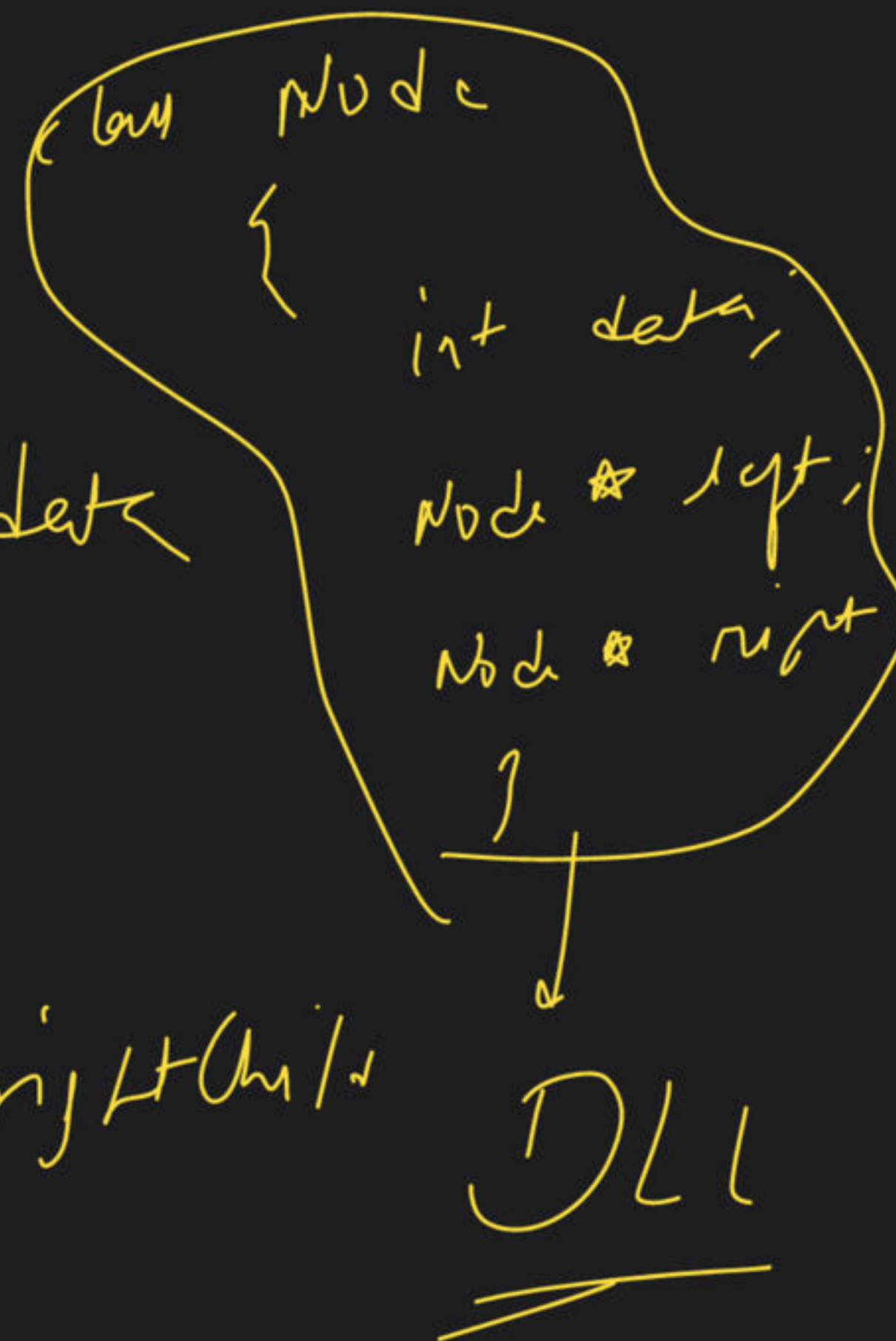


Node

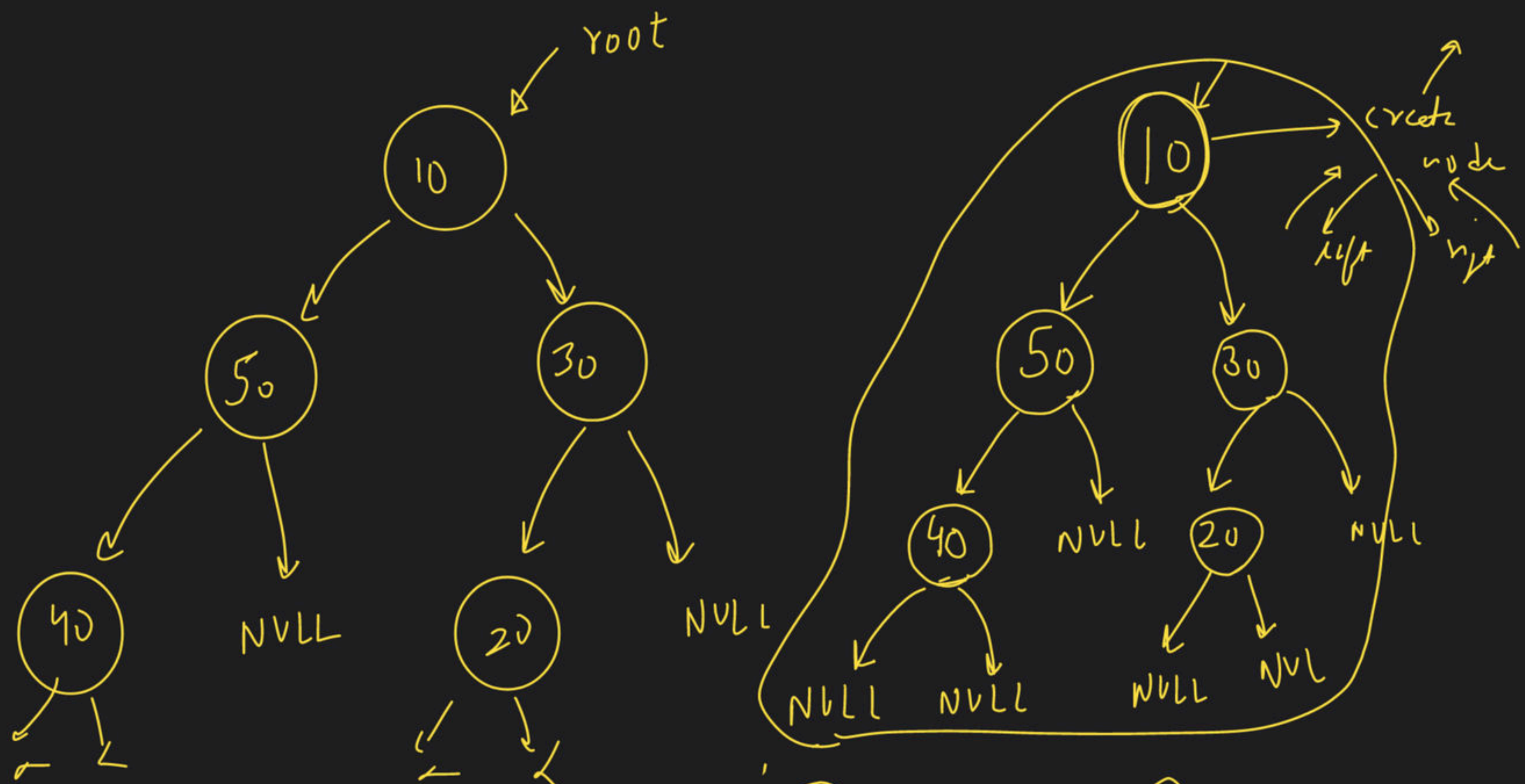
left child

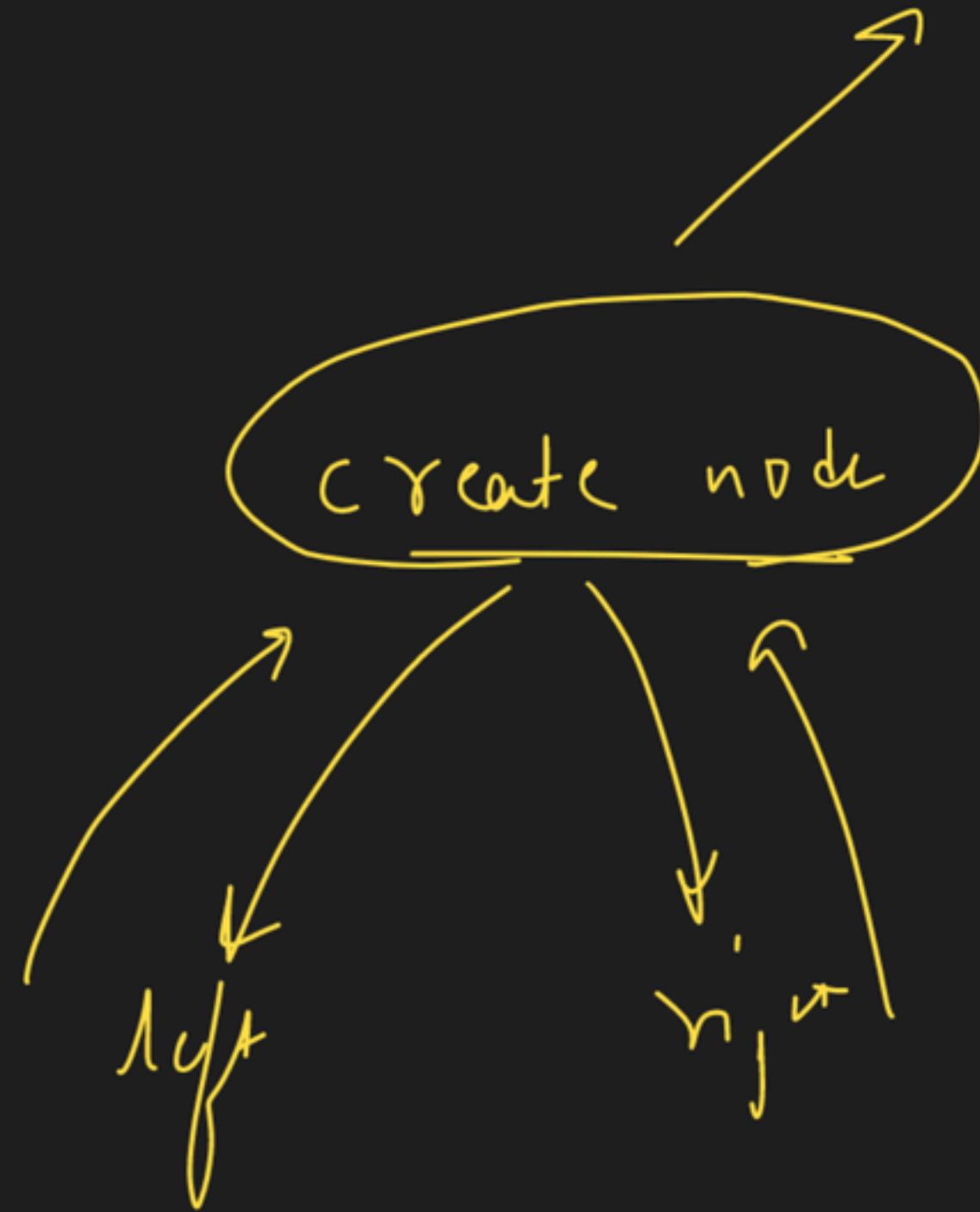


right child



→ Tree





10



2 min
Break

Traversal

99%

node
count

left subtree

right subtree

N L R

L N R

L R N

preorder

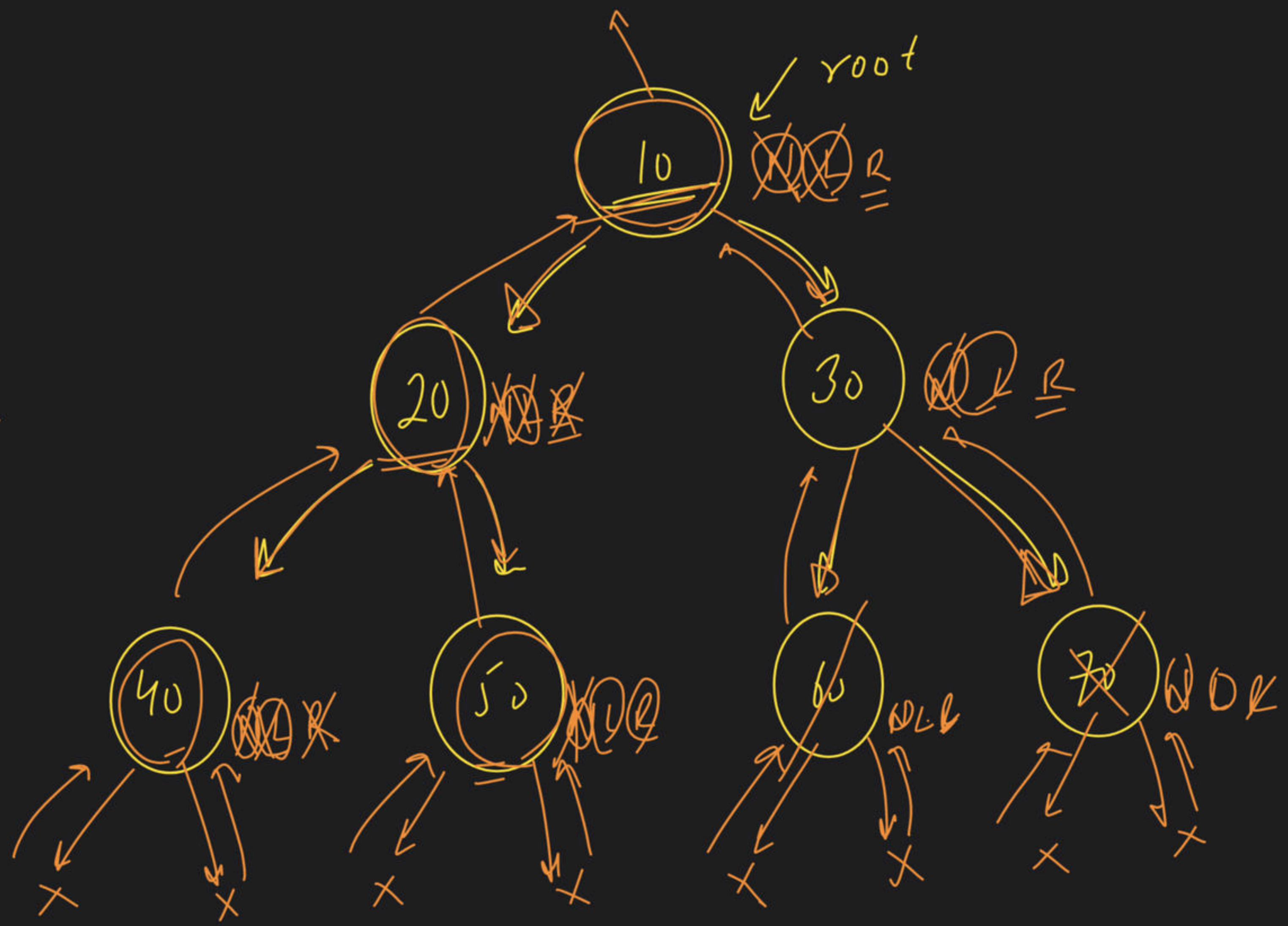
inorder

postorder

level order traversal

Morris traversal

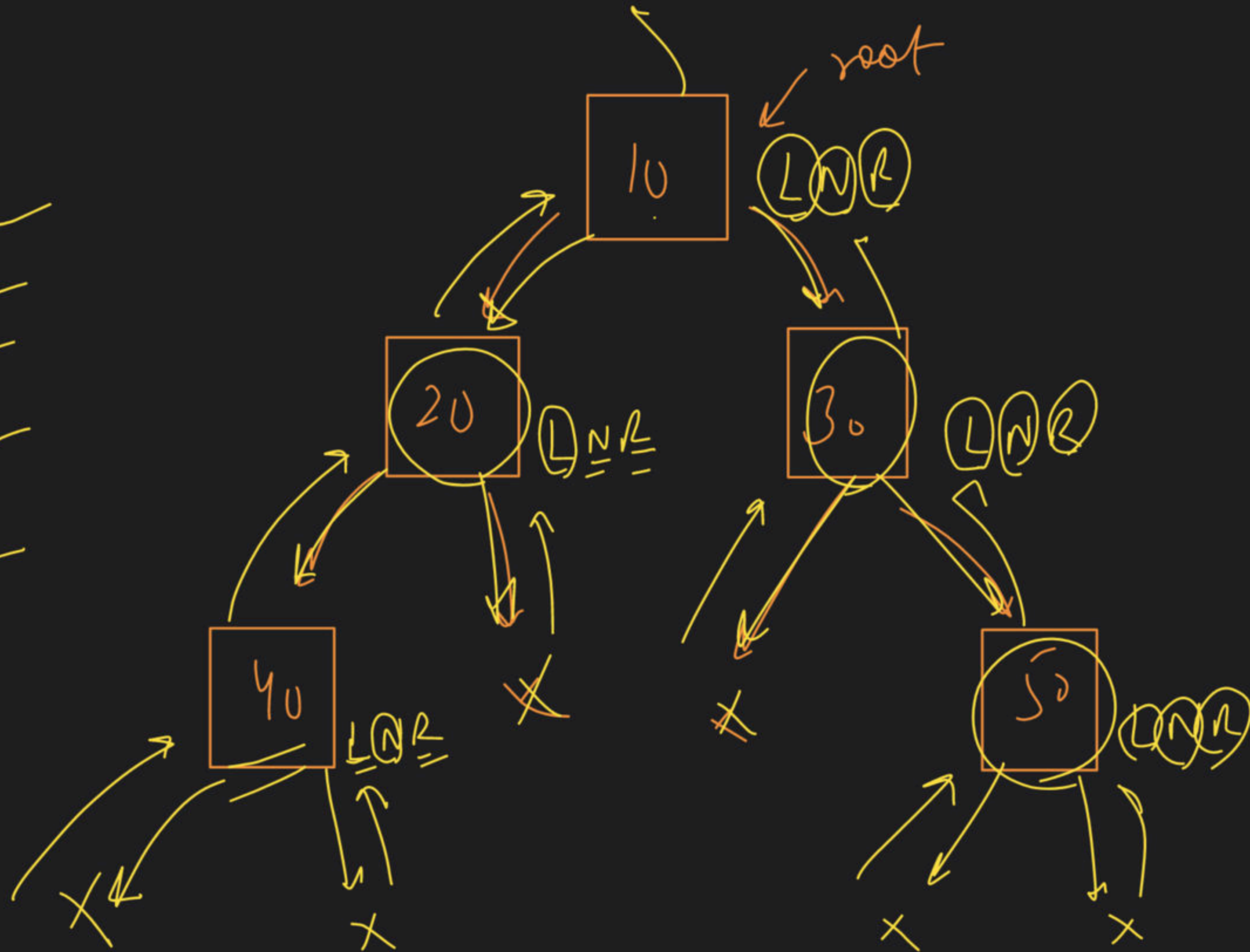
0/p
10-
20,
40
50-
30
60-
70/



preorder
NLR

NLR

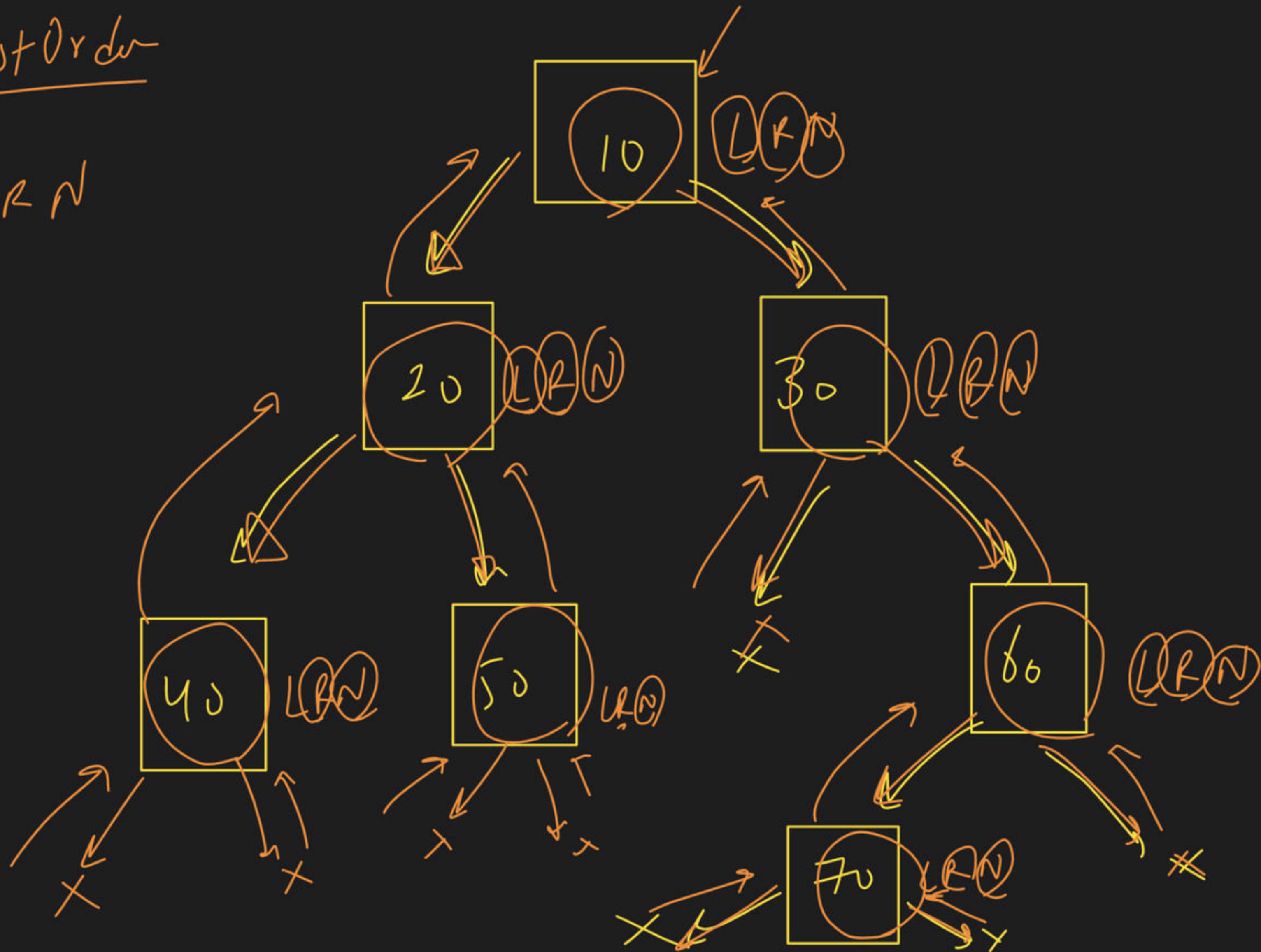
40 —
20 —
10 —
30 —
50 —



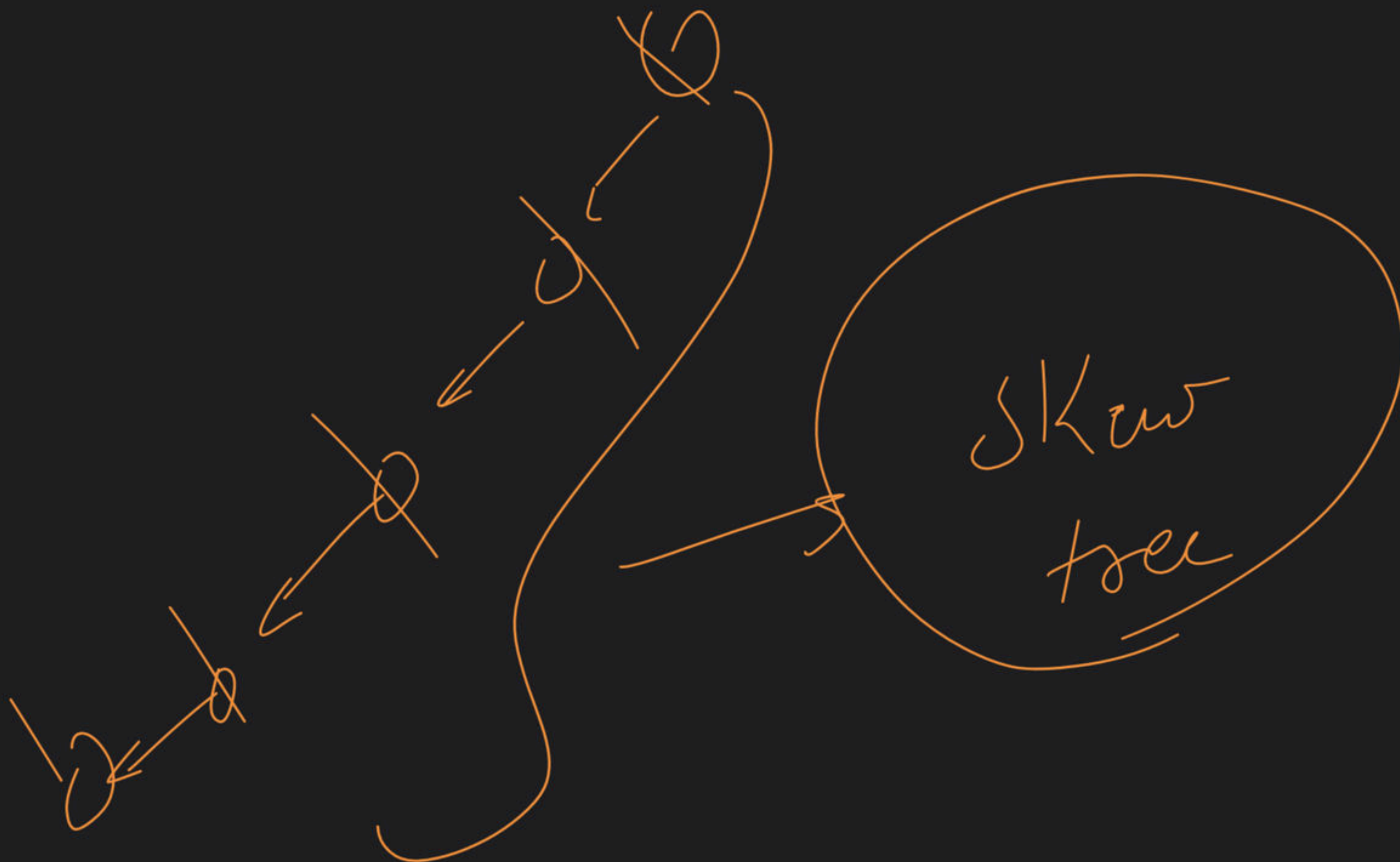
inorder
↓
LNR

PostOrder

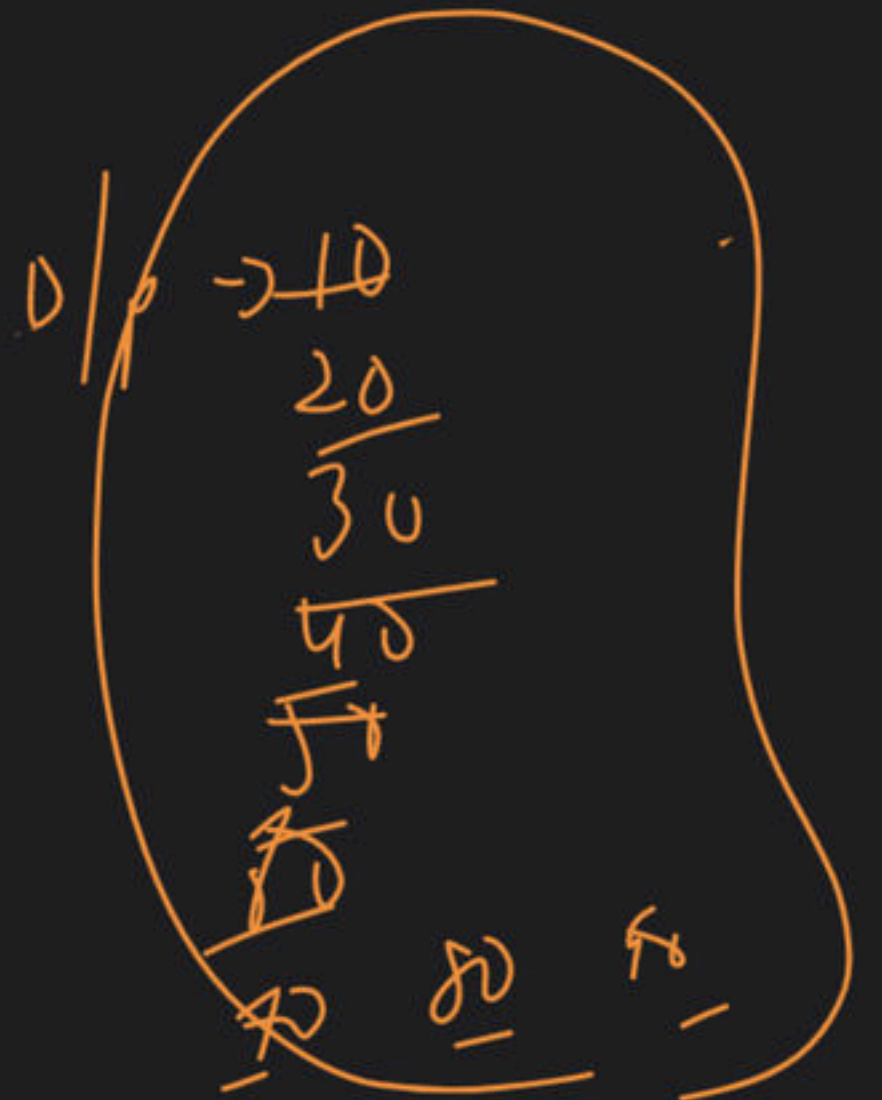
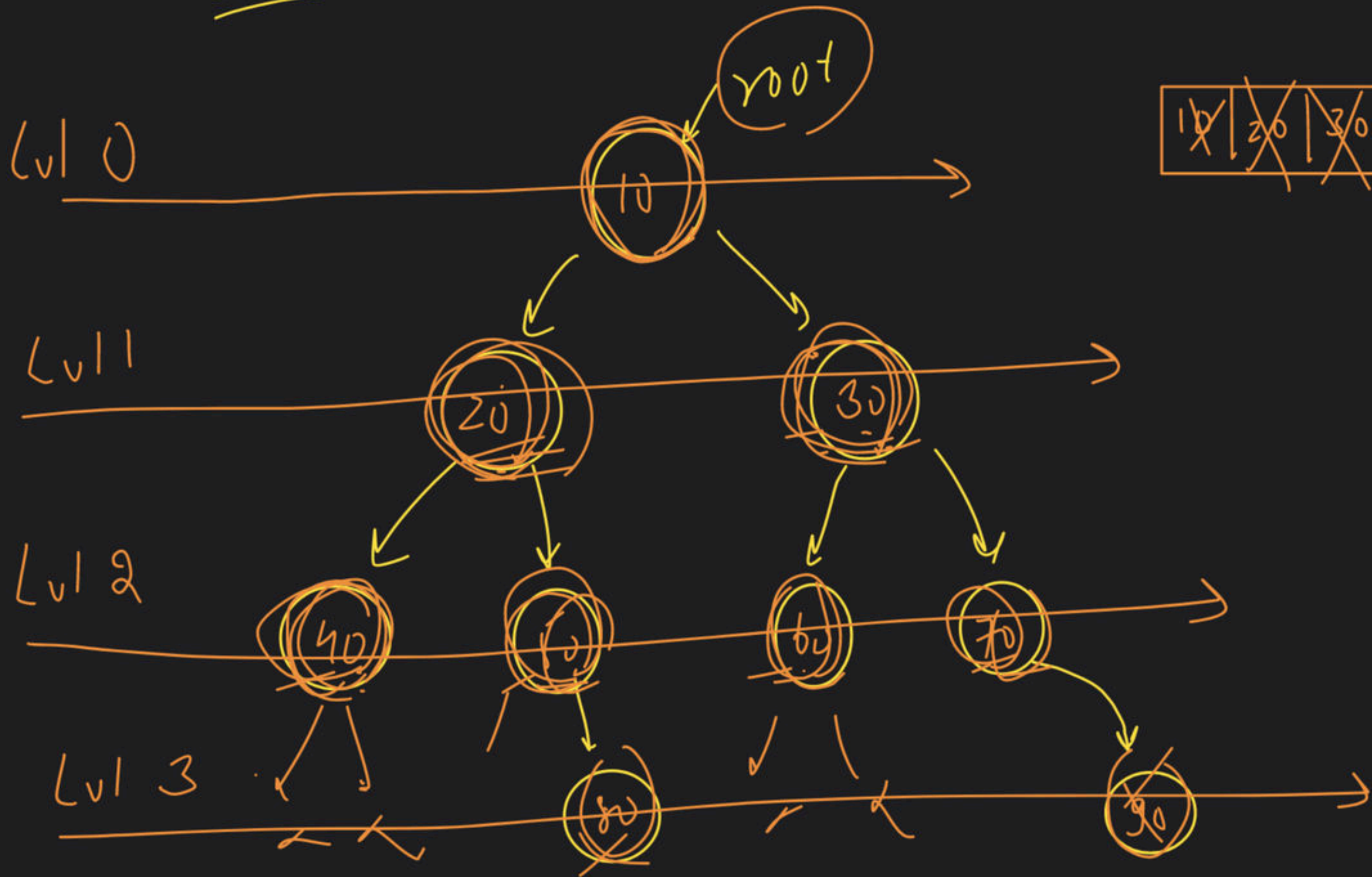
L R N



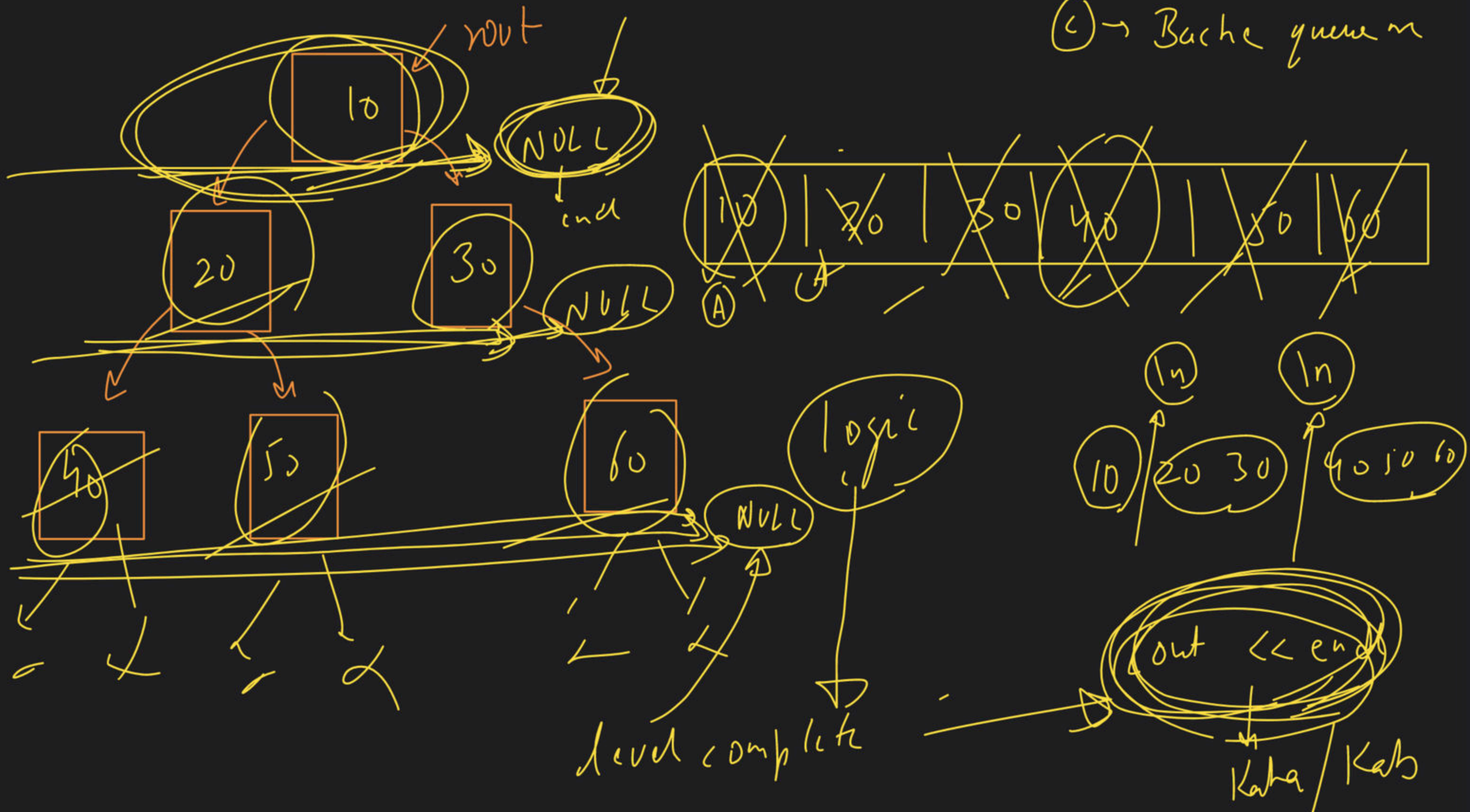
40
50
20
70
60
30
10

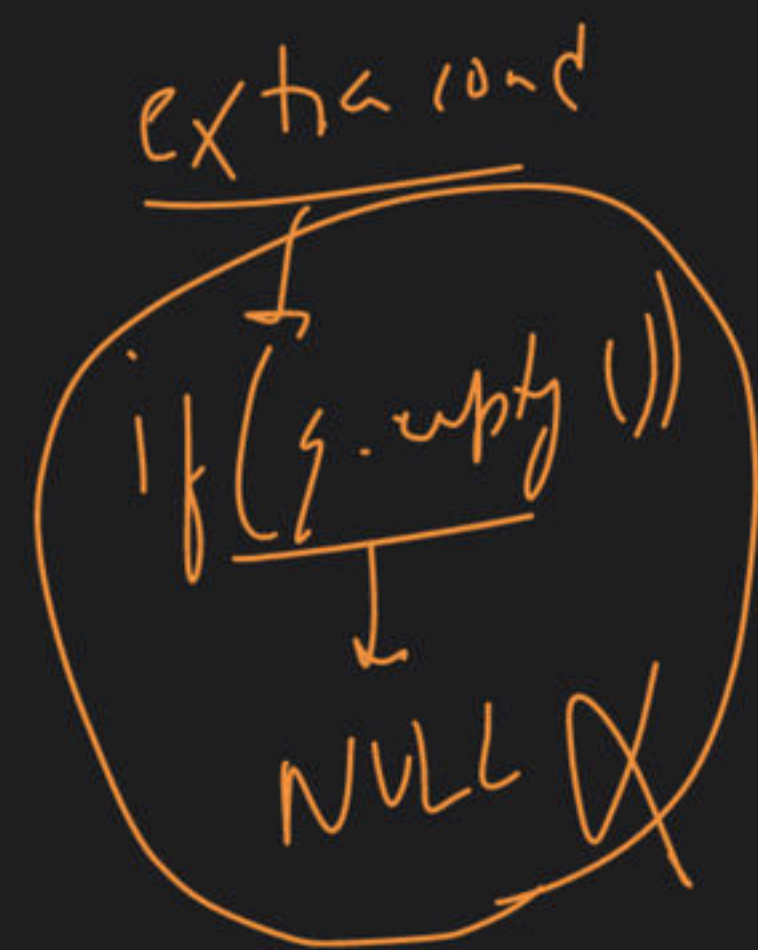
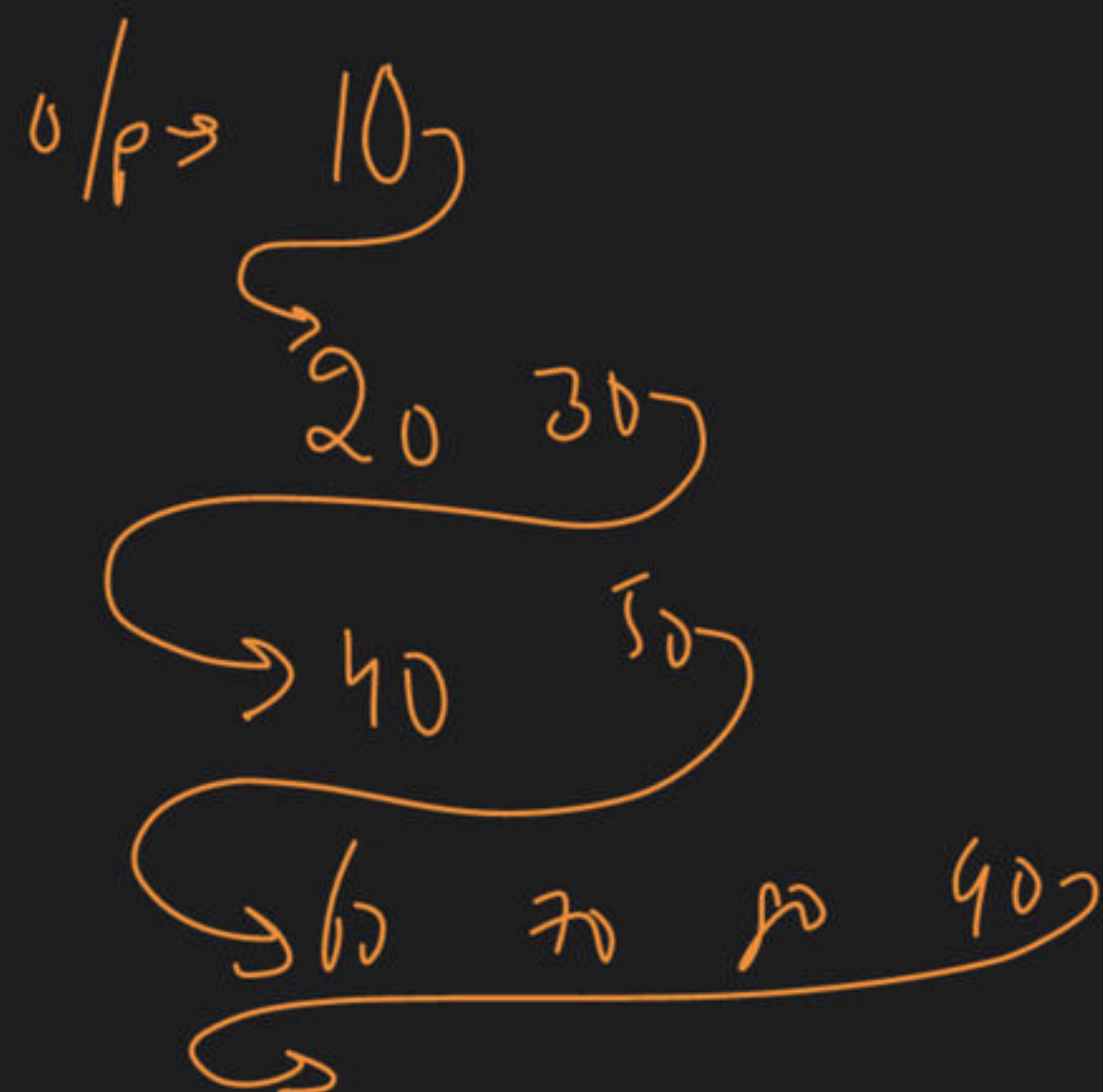
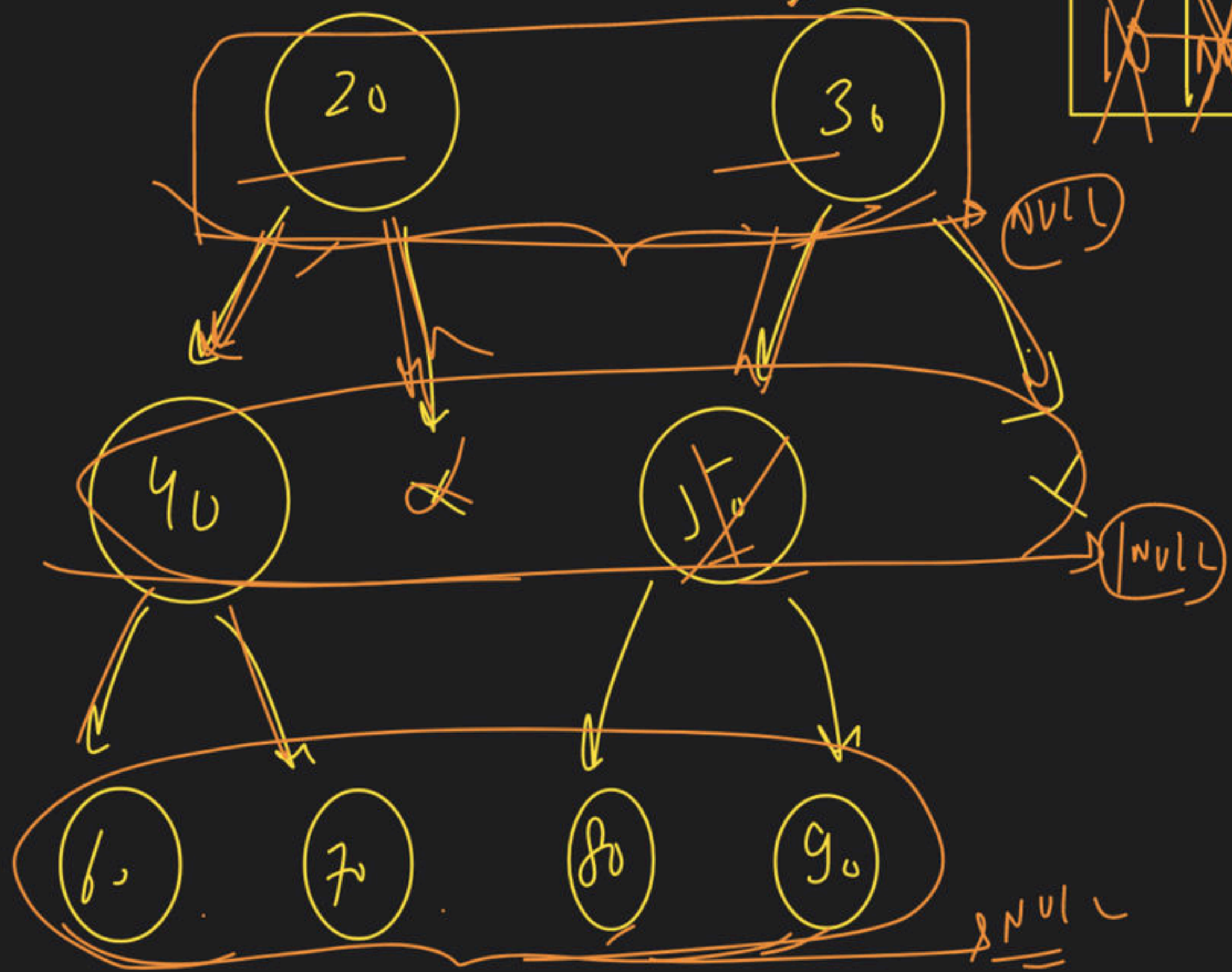
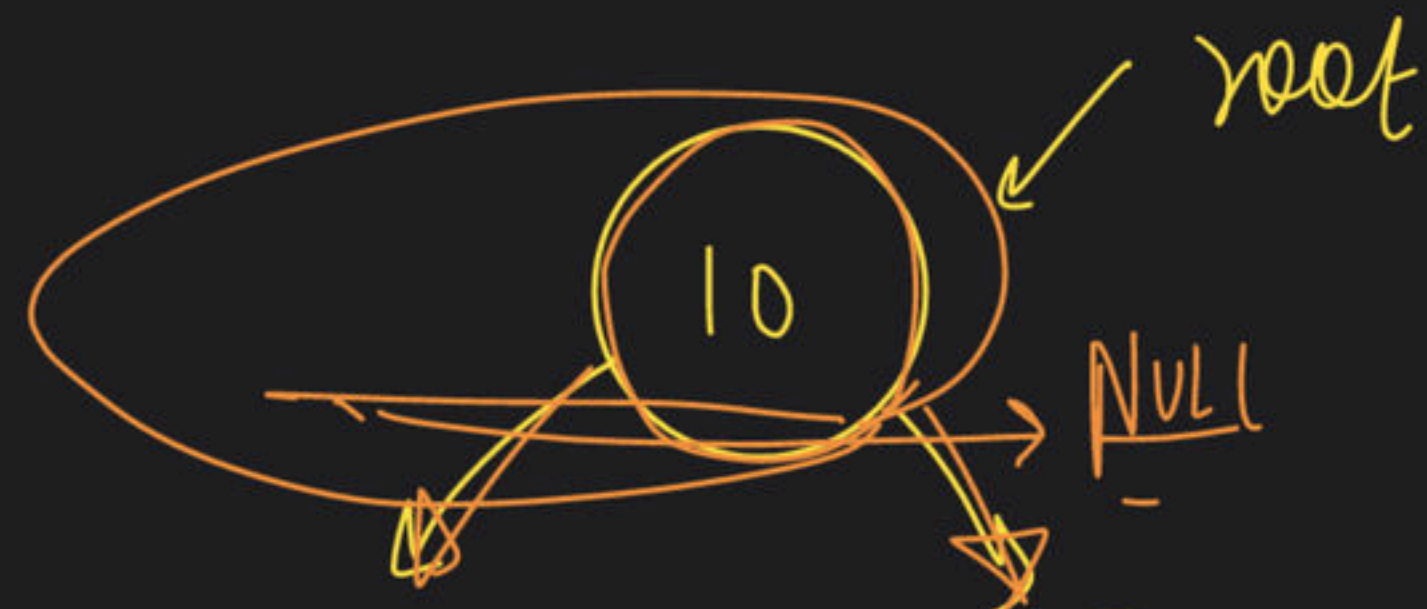


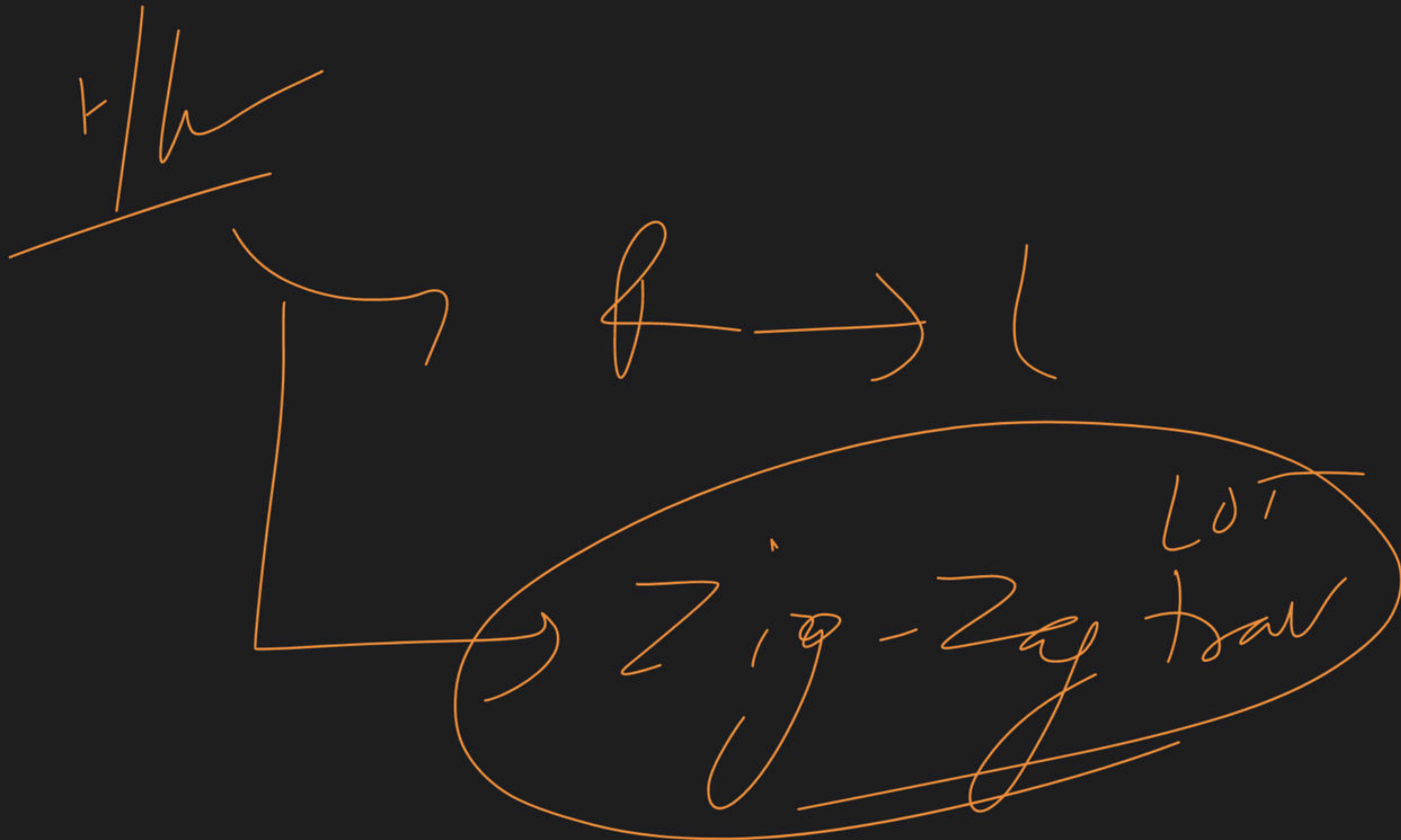
→ Level Order Traversal :-



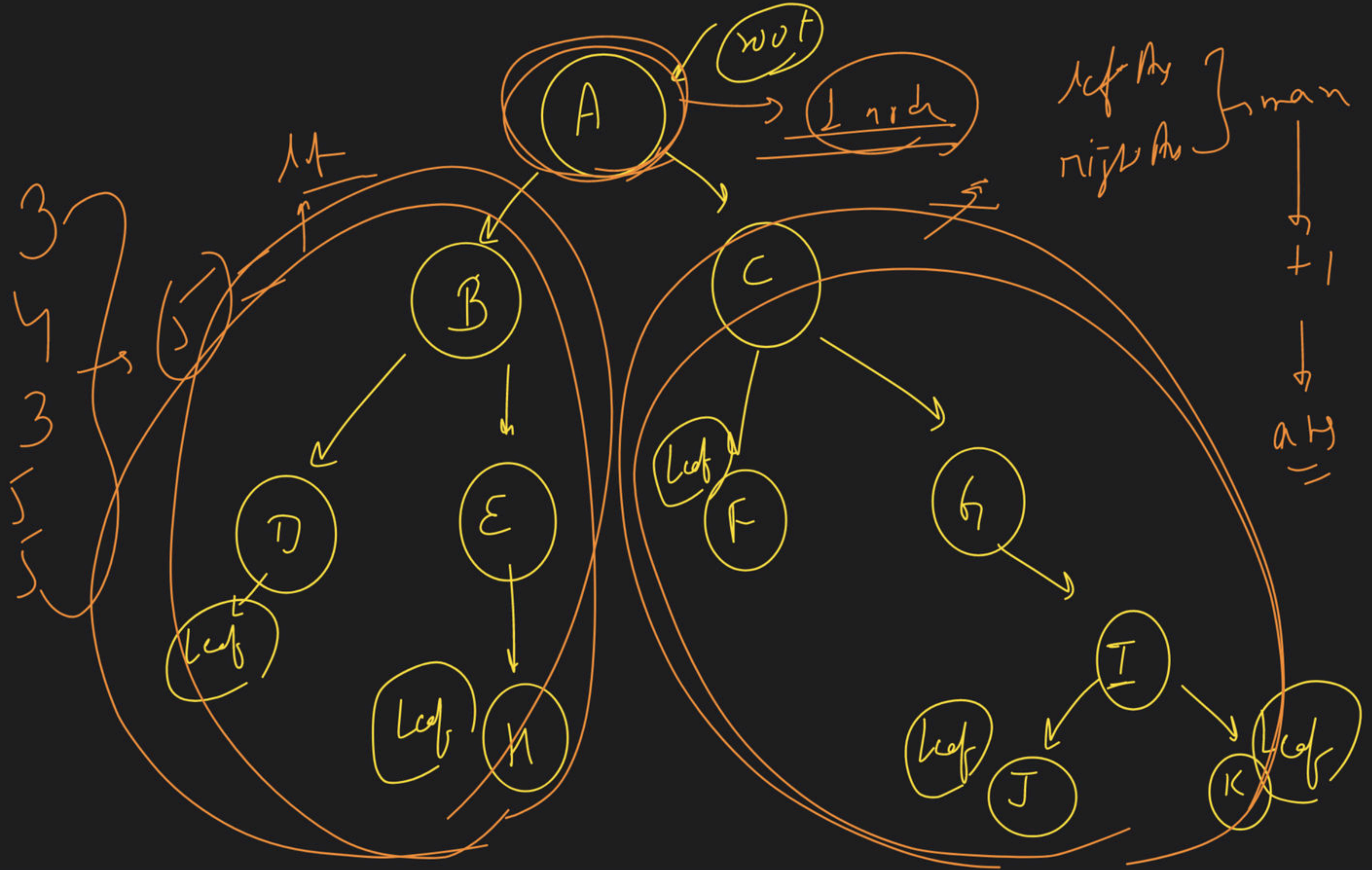
(c) → Bachhe gya m

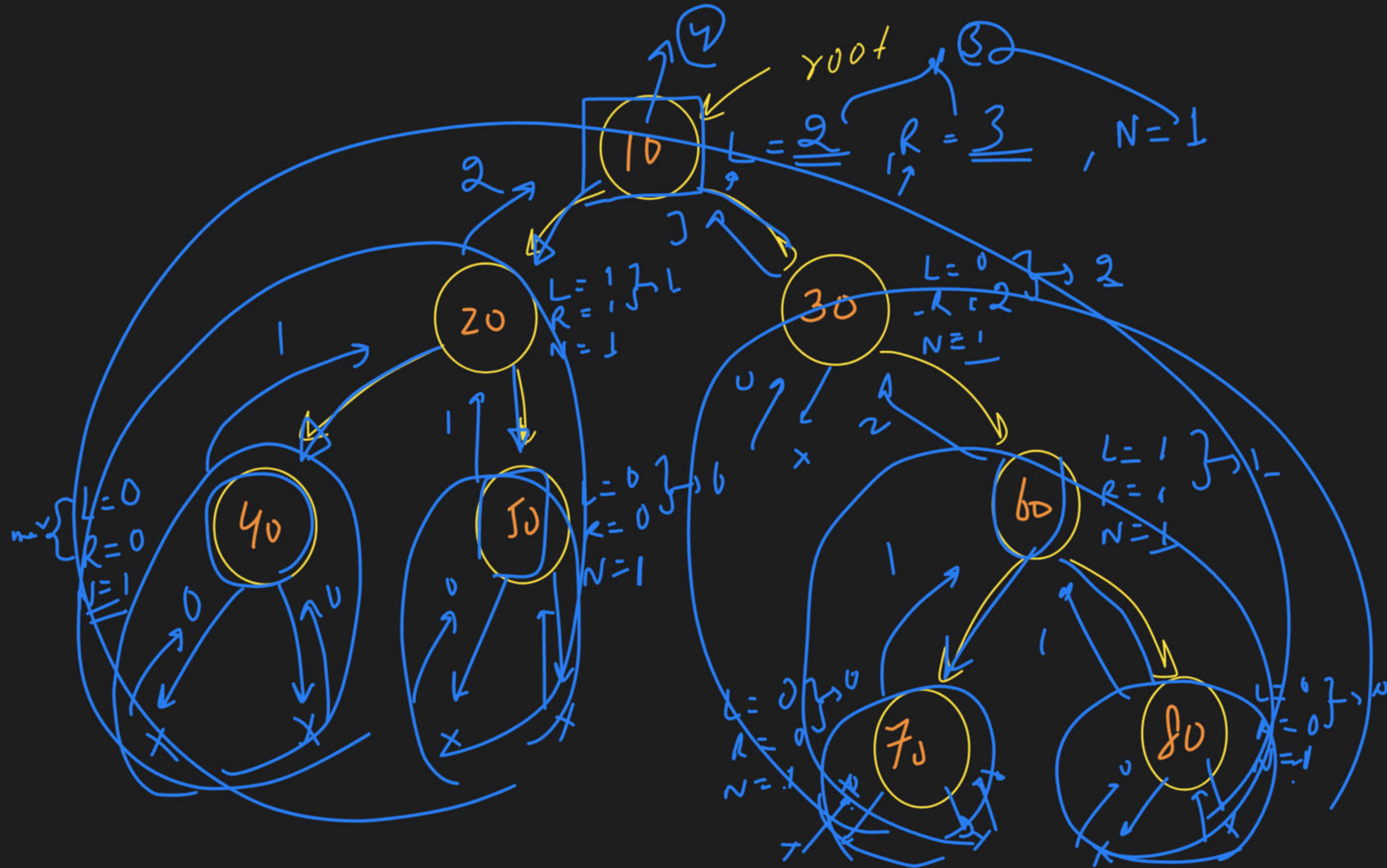


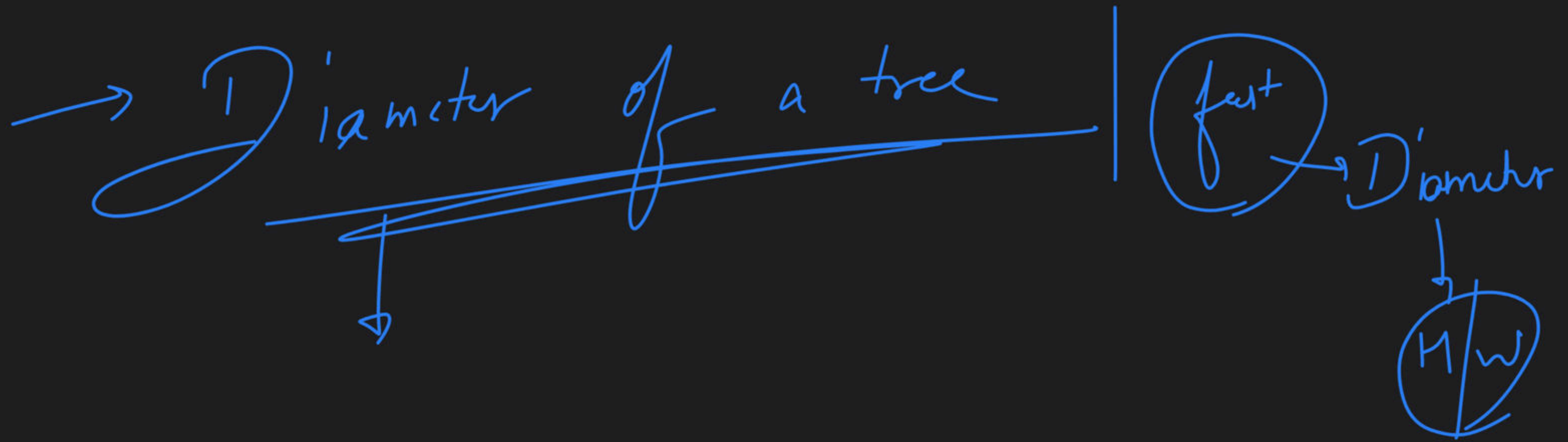


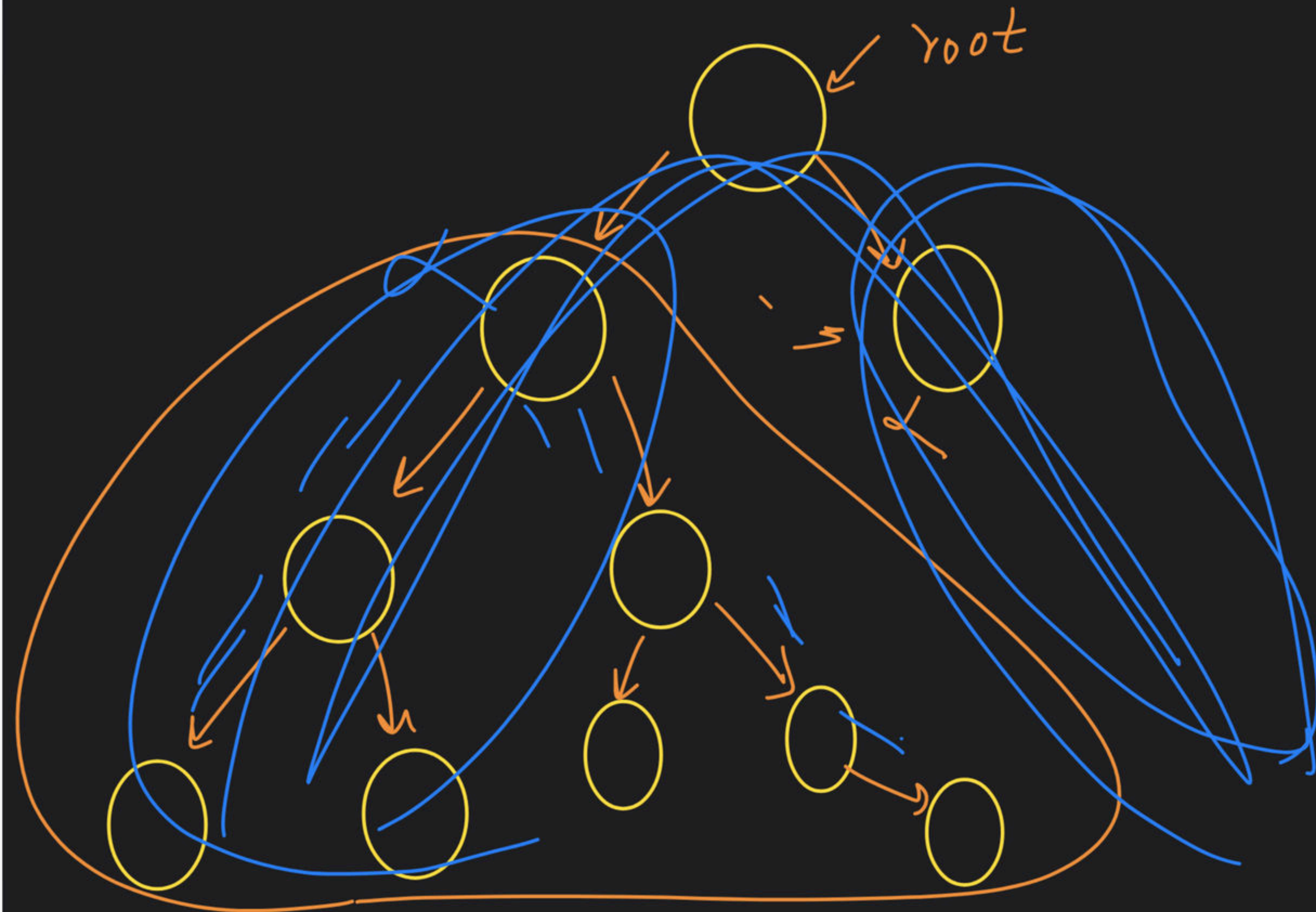


→ Height of Tree









normal

op $\rightarrow$ left subtree

op $\rightarrow$ right subtree

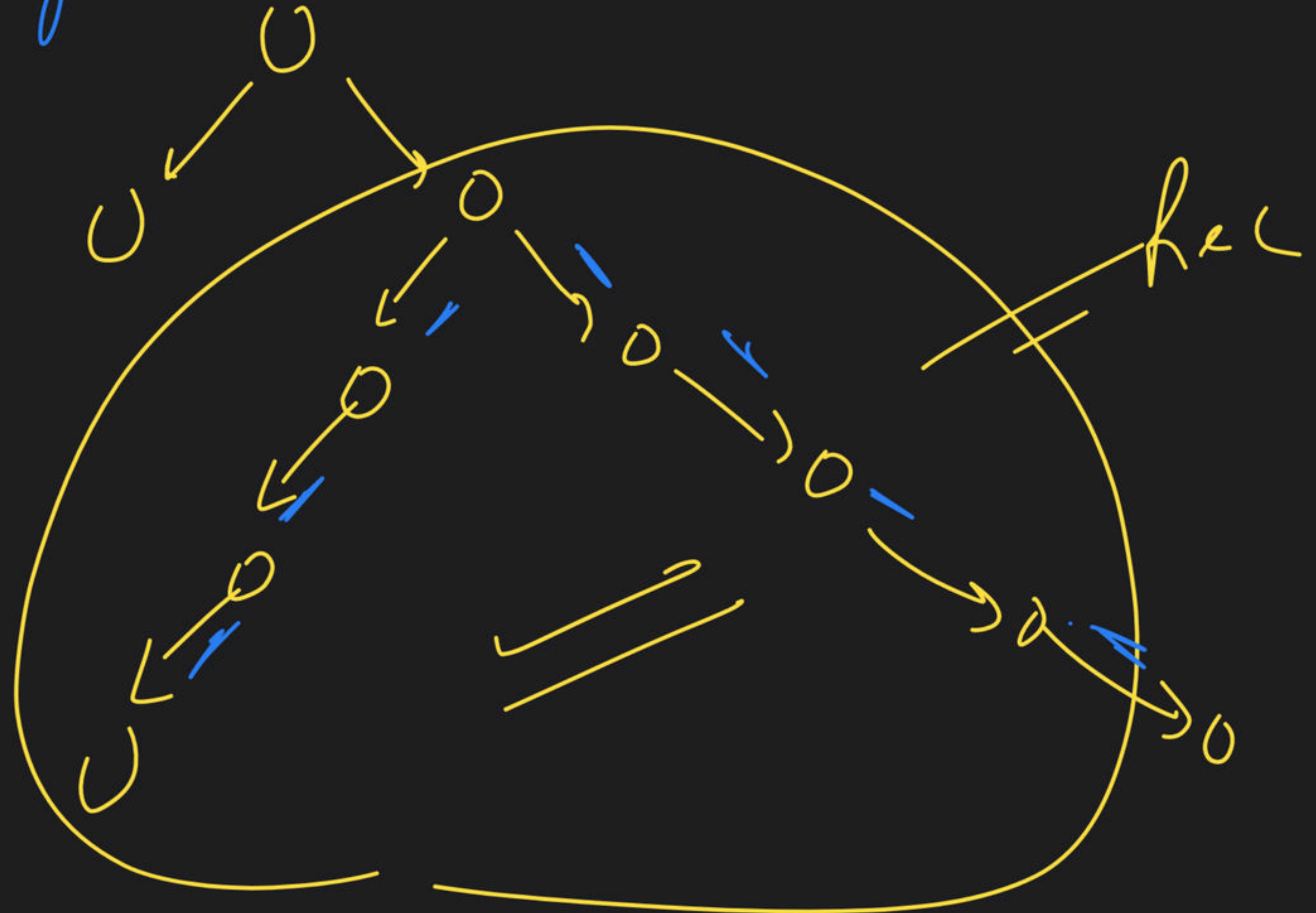
op $\rightarrow$ combine

op¹ → left subtree → ay



op2 →

right subtree → any



op3 $\rightarrow$ left & Right $\rightarrow$ ans $\rightarrow$ left+right
combination

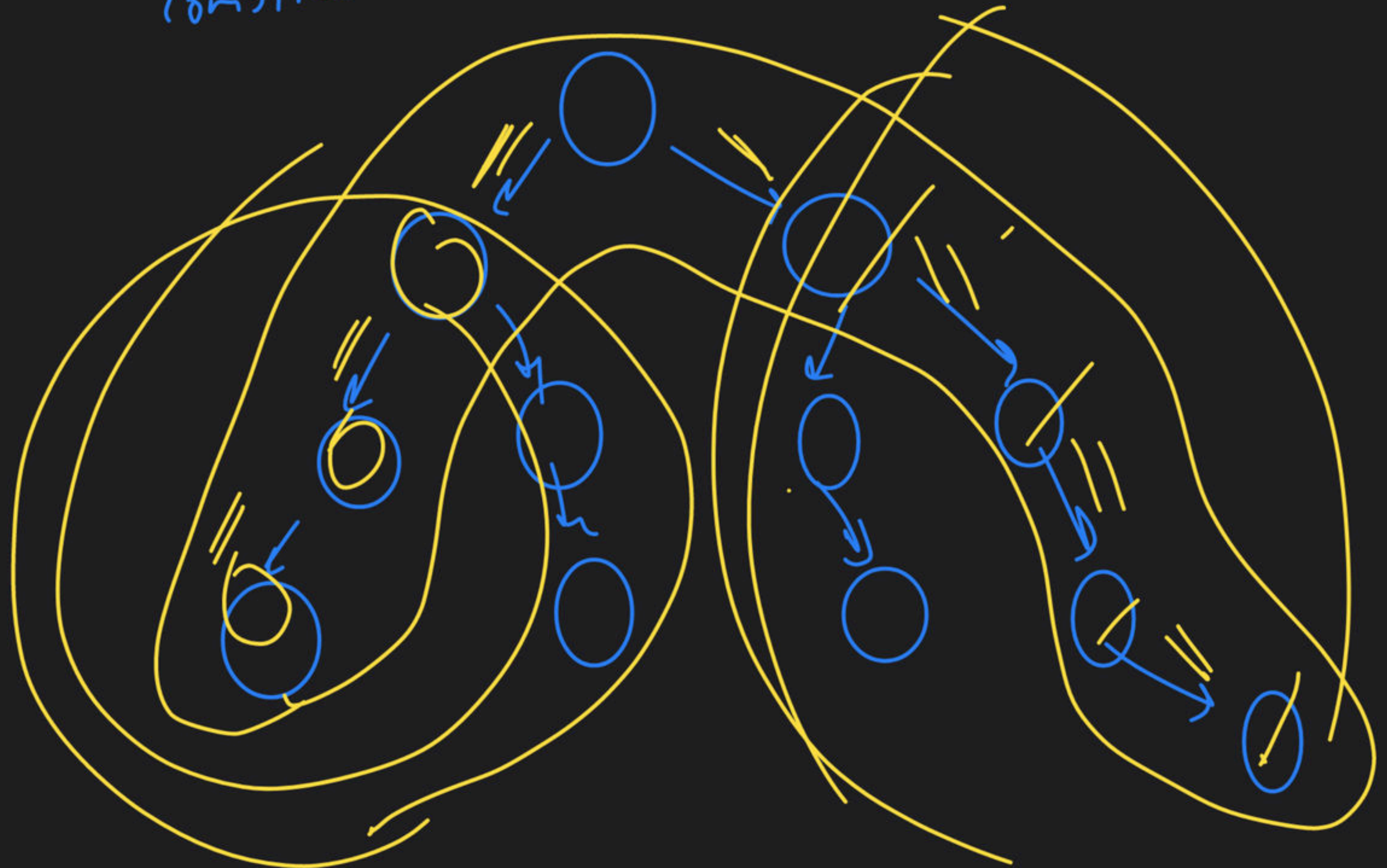


Diagram
pair

⑥

$op1 = 4$
 $op2 = 3$
 $op3 = 6$

root

$h \rightarrow$ height

$O(n^2)$

20

$op1 = 1$
 $op2 = 1$
 $op3 = 4$

30

$op1 = 0$
 $op2 = 1$
 $op3 = 2$

40

$op1 = 0$
 $op2 = 0$
 $op3 = 1$

50

$op1 = 0$
 $op2 = 0$
 $op3 = 1$

60

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

70

$op1 = 0$
 $op2 = 0$
 $op3 = 1$

80

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

90

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

95

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

int left 4

int Right 4

int $\rightarrow$ total $\rightarrow \max(L, R)$

?

+ 1

==

